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Senior Design I

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Individual Capstone Assessment

My group's senior design project is an application that combines many sports and esports in a pick 'em style bracket like fantasy football. Our idea is to weigh different games in a way that each sport is somewhat equal. To accomplish this, we will have a system to grant points for certain players getting a goal, making a pass, etc. Once we have the points allocated, we will set up a SQL database where we will hold all the data. Here, we will use the data for prediction numbers so anyone who doesn't have an idea as to how good a player is can have a general idea. Our application will be built via a website so users will be able to access our service from any device.

I have been preparing for a project like this from the classes: Data Structures CS 2028c, Database Design/Development CS 4092 and Programming Languages CS 3003. Data Structures gave me a general understanding of how different functions talk to each other within a given program or across multiple programs. Database Design/Development gave me more insight into SQL and how you can use complex queries to get sophisticated data sets. Despite already having experience with SQL from my first 2 co-ops, I wasn't using all the functionality of SQL that will be useful to this project. Programming Languages was practice of taking the skills and knowledge of other languages and leveraging them to quickly learn and integrate new languages into projects. Other computer science classes have also helped by teaching me common practice that can be used across all languages.

My first two co-op experiences were at Copeland as an Information Systems Co-op. Here I spent a lot of time working with SQL in a secure manner to limit the flow of information as needed. One thing I did there was create classifications for different people via their saved credentials so that only certain people could access certain information. This limited access to information will be critical to manage as we can't have users' information being disclosed to the public. My last three co-op terms were at a metal fabrication company called TMF. I mainly worked on some soft engineering work with CAM and AutoCAD software. While there, I made an automatic system to send emails when maintenance was due based on when the maintenance was last performed. It included some other things like materials needed and how much of those materials were left in stock. This furthered my understanding of how to pull data you need when you need it via rules you create.

I am excited to work on this project because it will need a lot of problem solving to make an exciting experience while still having balance between sports. Part of that process will be compiling data and finding a way to make each sport roughly equal in point balancing. The hardest part will be figuring out how to equate different markers for points in different sports. For example, comparing points in football and baseball is inherently challenging due to the scale of points: one run in baseball vs 6 points for a touchdown. To start the project, we will decide on one sport to create a pick 'em style bracket for. From here we can implement a website to host these capabilities. To populate the website with data, I will mainly work on the data server and connect queries to the website.

I expect us to have at least 2 sport leagues that players can be chosen from to get a full team. As the teams play and the data updates, points will be added to your team and whoever has the most points wins. After all the weeks of a season, the person with the most wins against other users wins. Evaluating work done may also be challenging, but since I am taking the lead on data management, I will implement queries for correct data. I will also use test cases to confirm that the queries are giving us correct data for the application of the queries. I want each of us to also run tests on work done by the others in the group so that we cover all our bases of bugs and functionality.